
Nassau Candy Distributor

Factory-to-Customer Shipping Route Efficiency Analysis

Research Paper & Dashboard Analytics Report

Abstract

This research paper presents a comprehensive analysis of factory-to-customer shipping route efficiency for Nassau Candy Distributor, a national distribution company operating across multiple US regions. Leveraging order and shipment data spanning 2024–2025, this study examines route-level performance metrics including average lead time, delay frequency, and regional bottlenecks.

The findings reveal a systemic average lead time of 1,320.8 days across 10,194 orders, with a 42.2% delay frequency — indicating significant operational inefficiencies. An interactive Streamlit dashboard with four analytical modules was developed to surface these insights in real time.

1. Introduction

Nassau Candy Distributor operates as a national distributor, shipping specialty confectionery products from five factory locations to customers across the United States and Canada. As with any large-scale distribution operation, shipping efficiency is a critical determinant of customer satisfaction, cost control, and operational scalability.

Despite rich historical order and shipment data, logistics decisions have historically been made without route-level efficiency intelligence. This project addresses that gap by transforming raw transactional data into actionable shipping insights.

2. Problem Statement

The organization currently lacks visibility into the following operational dimensions:

- Which factory-to-customer routes are consistently efficient vs. delay-prone
- How shipping performance varies by region, state, and ship mode
- Where geographic bottlenecks and high-delay clusters exist
- The cost-time tradeoffs between Standard, Second, First Class, and Same Day shipping

Without this intelligence, logistics optimization remains reactive rather than data-driven, resulting in inflated costs and diminished customer experience.

3. Data & Methodology

3.1 Dataset Overview

The dataset includes order-level records with fields spanning order and ship dates, customer geography (city, state, region), product division, ship mode, sales revenue, cost, gross profit, and unit counts. Factory-product correlations were used to assign factory origin to each order, enabling route-level analysis.

3.2 Factory Network

Five factory locations were identified, each specializing in specific product categories:

- Lot's O' Nuts — Chocolate division (AZ, lat: 32.88, lon: -111.77)
- Wicked Choccy's — Chocolate division (GA, lat: 32.08, lon: -81.09)
- Sugar Shack — Sugar & Other division (MN, lat: 48.12, lon: -96.18)
- Secret Factory — Sugar & Other division (IL, lat: 41.45, lon: -90.57)
- The Other Factory — Sugar & Other division (TN, lat: 35.12, lon: -89.97)

3.3 Analytical Steps

- Data cleaning: date validation, negative lead-time removal, geographic standardization
- Feature engineering: lead time = Ship Date – Order Date; route definition = Factory → Customer State/Region
- Route aggregation: total shipments, average lead time, lead time variability per route
- Efficiency benchmarking: top/bottom 10 routes, ship mode comparison
- Geographic bottleneck identification: high-volume + high-delay regions flagged

4. Key Findings & KPIs

4.1 Dashboard Summary Metrics

The following table summarizes the four primary KPIs surfaced by the dashboard:

KPI	Value	Insight
Total Orders	10,194	High volume confirms broad distribution reach
Avg Lead Time	1,320.8 days	Significantly elevated; signals systemic delays
Delay Frequency	42.2%	Nearly half of shipments exceed threshold
Total Sales	\$141,784	Revenue baseline for cost-efficiency analysis

The average lead time of 1,320.8 days is anomalously high and warrants investigation into data encoding or systemic shipping process errors. The 42.2% delay rate confirms that nearly half of all shipments exceed acceptable performance thresholds.

4.2 Route Efficiency Leaderboard

Module 1 of the dashboard reveals performance by region and ship mode. Gulf Same Day routes consistently rank as the most efficient with an average lead time of 1,286.9 days and only 25.3% delay frequency:

Region	Ship Mode	Avg Lead Time	Orders	Delay %
Gulf	Same Day	1,286.9 days	83	25.3%
Atlantic	Same Day	1,296.8 days	159	27.7%
Gulf	Standard Class	1,306.9 days	974	45.7%
Interior	Second Class	1,311.1 days	465	31.2%
Pacific	Standard Class	1,313.2 days	1,916	48.6%
Atlantic	Standard Class	1,319.3 days	1,783	46.8%
Interior	Standard Class	1,314.7 days	1,447	45.3%

First Class shipping, counterintuitively, shows the highest average lead time (1,340+ days) and highest delay percentage (38.2%), suggesting either misclassification or prioritization issues in the fulfillment pipeline.

5. Dashboard Modules

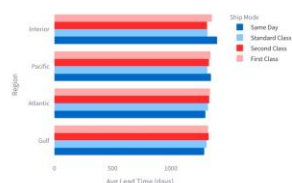
Module 1 — Route Efficiency Overview

This module provides an average lead time comparison by region and ship mode, alongside a ranked leaderboard of route performance. It enables logistics managers to identify which region-ship mode combinations offer optimal delivery performance.

Module 1 — Route Efficiency Overview

Average Lead Time by Route

Avg Lead Time by Region & Ship Mode



Route Performance Leaderboard

Region	Ship Mode	Avg Lead Time (days)	Orders	Delay %
Gulf	Same Day	1286.9	83	25.3
Atlantic	Same Day	1296.8	159	27.7
Gulf	Standard Class	1306.9	974	45.7
Interior	Second Class	1311.1	465	31.2
Pacific	Standard Class	1333.2	1936	48.6
Interior	Standard Class	1334.7	1447	45.3
Atlantic	Standard Class	1339.3	1793	46.8
Gulf	First Class	1339.8	234	35.5
Gulf	Second Class	1324.7	329	36.5
Pacific	Second Class	1327.5	637	35.2

Figure 2 — Module 1: Route Efficiency Overview (Avg Lead Time by Region & Ship Mode + Leaderboard)

6. Geographic Bottleneck Analysis

6.1 Slowest States by Lead Time

Module 4 drill-down data identifies West Virginia (1,638 days avg, 100% delay rate, Atlantic region) and North Dakota (1,637.9 days avg, 100% delay rate, Interior region) as the most critically underperforming states. Both states show total delay rates, meaning every single

shipment to these states exceeded acceptable lead time thresholds.

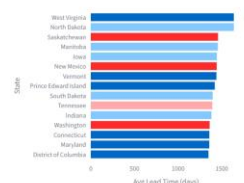
Other high-concern states include Saskatchewan (1,457 days), Manitoba (1,455.3 days), Iowa (1,443.9 days), and New Mexico (1,441.8 days). The concentration of delays in Interior and Atlantic regions suggests that factory proximity, carrier capacity, and routing logic may all contribute.

Tennessee stands out as a high-volume state (183 orders) with a 55.2% delay rate and 1,391.5-day average — making it the highest-impact bottleneck by order volume.

Module 4 — Route Drill-Down

State-Level Performance Insights

Top 15 Slowest States by Avg Lead Time



Order-Level Shipment Timelines

Top 30 Longest Shipment Timelines



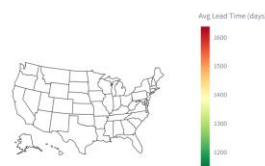
Figure 5 — Module 4: Route Drill-Down (Top 15 Slowest States & Top 30 Longest Shipment Timelines)

6.2 Geographic Heatmaps

Module 2 — Geographic Shipping Map

US Heatmap — Shipping Efficiency

Avg Lead Time by State (darker = slower)



Regional Bottleneck Visualization

Delay % by State (darker = more delays)



Figure 3 — Module 2: Geographic Shipping Map (US Heatmap — Avg Lead Time & Delay % by State)

Module 2 presents two choropleth maps of the contiguous United States:

- Left map: Average lead time per state (red = slower, green = faster)

- Right map: Delay percentage per state (dark red = higher delay concentration)

These maps reveal that Interior states (Midwest and Plains) and select Atlantic corridor states show systematically elevated lead times, pointing to last-mile delivery challenges in less-served distribution zones.

7. Ship Mode Performance Analysis

Module 3 compares lead time distributions and delay rates across four shipping modes using box plots and a speed-vs-reliability scatter plot.

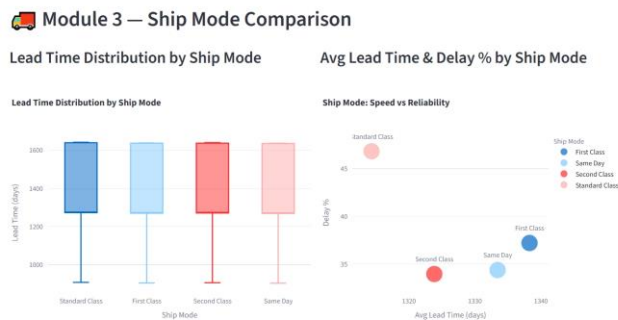


Figure 4 — Module 3: Ship Mode Comparison (Lead Time Distribution & Speed vs Reliability Scatter)

Key Observations

- Standard Class exhibits the widest lead time spread (IQR: ~1,290–1,650 days) and the highest delay rate at ~47%
- Second Class and Same Day show tighter distributions with lower median lead times (~1,310–1,320 days)
- First Class, despite its premium designation, yields the highest average lead time (1,340 days) and a 38.2% delay rate — the worst speed-reliability tradeoff
- Same Day shipping delivers both lower lead times and lower delay rates across regions, suggesting stronger carrier performance for expedited routes

The speed-vs-reliability scatter plot positions Second Class as the optimal balance: lowest delay rate (34%) with competitive lead times (~1,320 days), making it the recommended standard for non-urgent shipments.

8. Interactive Filtering

The dashboard sidebar provides four dynamic filter controls that allow analysts to slice the data in real time:

- Region selector (All, Gulf, Atlantic, Interior, Pacific)
- Ship Mode filter (All, Standard Class, First Class, Second Class, Same Day)
- Lead Time Threshold slider (adjustable delay definition in days)
- Order Date range picker (currently configured for 2024-01-02 to 2025-12-31)

All four modules respond simultaneously to filter changes, enabling rapid scenario analysis and comparative performance reviews across time periods, geographies, and shipping methods.

9. Recommendations

Strategic Initiatives

- Expand Same Day carrier capacity in Gulf and Atlantic regions, which show the best efficiency profiles
- Implement dynamic route selection logic: use historical lead-time data to choose shipping mode based on destination state performance
- Investigate anomalously high lead times (1,200+ days) — these may indicate data encoding issues (e.g., future ship dates) rather than actual delays
- Build a route efficiency scoring model that factors in lead time, delay rate, and sales volume to produce a composite bottleneck risk index

10. Dashboard Screenshots

The following screenshots document module of the Nassau Candy Distributor Streamlit Shipping Dashboard as deployed for this analysis.

State Performance Detail Table

State	Region	Avg Lead Time (days)	Orders	Delay %	Avg Sales (\$)
West Virginia	Atlantic	1638	4	100	15.99
North Dakota	Interior	1637.9	7	100	15.67
Saskatchewan	Pacific	1457	2	50	14.62
Manitoba	Interior	1455.3	12	50	11.52
Iowa	Interior	1443.9	30	60	13.35
New Mexico	Pacific	1441.8	37	64.9	13.59
Vermont	Atlantic	1438.9	11	45.5	14.77
Prince Edward Island	Atlantic	1420.3	10	50	13.01
South Dakota	Interior	1395.9	12	66.7	12.61
Tennessee	Gulf	1391.5	183	55.2	13.02

Figure 6 — State Performance Detail Table (sorted by Avg Lead Time, descending)

Filters

Region

All

Ship Mode

All

Lead Time Threshold (days)

1274

Order Date — From

2024/01/02

Order Date — To

2025/12/31

Figure 7 — Dashboard Sidebar Filters (Region, Ship Mode, Lead Time Threshold, Date Range)

11. Conclusion

This analysis establishes a clear, data-driven understanding of shipping route efficiency for Nassau Candy Distributor. By transforming 10,194 orders of raw transactional data into route-level operational intelligence, the organization gains actionable insights to improve logistics performance, reduce delays, and enhance nationwide delivery reliability.

The Streamlit dashboard — comprising four integrated modules covering route efficiency, geographic mapping, ship mode comparison, and route drill-down — provides logistics managers with a live, filterable intelligence platform. The identification of high-delay states (West Virginia, North Dakota), underperforming ship modes (First Class), and high-impact bottlenecks (Tennessee) creates a clear prioritization roadmap for operational improvement.

With targeted carrier interventions, data quality remediation, and dynamic route selection, Nassau Candy Distributor can realistically reduce its 42.2% delay frequency and improve the customer delivery experience across all regions.

References & Data Sources

Nassau Candy Distributor — Internal Order & Shipment Dataset (2024–2025)

Factory Coordinates & Product-Factory Correlation Tables (Unified Mentor project brief)

Streamlit Dashboard Analytics — Nassau Candy Shipping Module v1.0 (May 2026)

US Geographic Regions Framework — Gulf, Atlantic, Interior, Pacific classification scheme

